

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

The factors affecting the pricing policies of Fish-Water Farm are:

1. Cost of production - direct and indirect costs, variable and fixed costs, and mixed costs will all directly impact the price setting.
2. Demand and competition in the market - higher prices can only be set for those products with high demand or differentiated by special features in the market.
3. Consumer psychology - psychological pricing refers to the practice of setting prices slightly lower than rounded numbers. This is based on the idea that customers do not round up and will treat them as lower prices than they actually are.
4. Political and legal aspects - government policy on price control with a price ceiling or price floor for goods and services provided will also influence the price setting of a product.

Answer 1(b)

The unit cost for each 100g salad box sold by Fish-Water Farm is HK\$65.5 with calculations below:

	HK\$
Direct labour costs	2,950,000
Fish feed and seeds supplies	1,380,000
Water	2,780,000
Electricity	3,698,000
Transportation	738,000
Veterinary and hygiene	810,000
Fixed overheads	8,056,000
Depreciation	1,894,000
Sundry expenses	1,286,000
	23,592,000

$$\text{HK\$}23,592,000 \div 360,000 = \text{HK\$}65.5$$

*The cost of building the new indoor aquaponics farm is a sunk cost which is irrelevant to the unit cost calculation.

Answer 1(c)

Fish-Water Farm is adopting "market penetration pricing" - by charging a unit selling price of HK\$58, which is deliberately set below its unit cost of production of HK\$65.5 for each salad box sold.

Market penetration pricing can help companies boost their market share and sales volume. In turn, a higher sales volume can lead to lower production costs (fixed costs per unit) and higher profits. A low price set as part of an introductory campaign may entice customers to switch from traditional products to the new products of the company.

Answer 1(d)

The public education activities can be regarded as advertisements to build a good reputation for Fish-Water Farm.

The free green workshops and site visits arranged by Fish-Water Farm are regarded as corporate social responsibility (CSR) engaged by the company to create a positive impact on the community that contributes to the overall wellness of society and to act as a good corporate citizen.

Fish-Water Farm can make use of the free workshops and site visits to promote its products and increase its sales turnover by offering different promotional packages, such as:

- Adopting a complementary product pricing strategy by setting lower prices for some products to achieve higher sales in order to increase demand for other products that have higher profit margins;
- A bulk purchase discount;
- A discount for the next order on the online shop;
- A membership discount to increase customer loyalty to the company's products; and
- Adopting a price discrimination strategy by charging different prices to different groups of customers (general public, elderly, and school children, etc.) when buying the same products.

Answer 2(a)

Free cash flow for Year 3	HK\$
Sales revenue	5,500,000
Less:	
Cost of goods sold (20%)	(1,100,000)
Operating expenses	(620,000)
Depreciation	(215,000)
Income before interest and tax	3,565,000
Less: Taxation (20%)	(713,000)
Net operating profit after tax	2,852,000
Add: Depreciation	215,000
Cash flow from operations	3,067,000
Less:	
Capital expenditures	(600,000)
Additions to working capital	(250,000)
Free cash flows	2,217,000

Firm value of the electronic cigarette company		<u>WACC</u> <u>(10%)</u>	PV
	HK\$		HK\$
Free cash flow for Year 1	1,500,000	0.1	1,363,636.36
Free cash flow for Year 2	2,000,000	0.1	1,652,892.56
Free cash flow for Year 3	2,217,000	0.1	1,665,664.91
Terminal value after year 3	*58,750,500	0.1	44,140,120.20
Firm value			<u>48,822,314.03</u>

*Terminal value after year 3 = HK\$2,217,000 x 1.06 / (0.1 – 0.06)

Answer 2(b)

Firm value of the hand-crafted beer company:

HK\$5.83 x 10.25 x 1,000,000 shares = HK\$59,757,500

Answer 2(c)

Apart from the financial evaluations, there are other non-financial considerations for the investment appraisal as listed below:

- The alcohol laws of Hong Kong and the sale of alcohol are relatively liberal and lenient in the sense that alcohol can be provided by any licensed restaurants, bars, clubs and food retailers (including supermarkets), subject to restrictions related to the legal drinking age of at least 18. However, Urban Park has to apply for the license to sell alcohol and must to be aware of the selling restrictions.
- Some consumers may find pleasure and enjoyment when drinking wine and/ or hand-crafted beers with friends or at family gatherings. Consumers who drink may order more food, which will boost overall sales revenue.

Answer 3(a)

	<u>Vertex</u>	<u>Peak</u>
<u>Liquidity:</u>		
Current ratio	1.94	1.52
Quick ratio	1.64	1.14
<u>Asset management efficiency:</u>		
Inventory turnover	5.90	4.87
Days' receivables	16.98	20.55
Total asset turnover	1.21	1.18

Leverage:

Debt ratio	0.57	0.35
Equity multiplier	2.35	1.53
Times Interest Earned	3.92	3.41

Profitability:

Net profit margin	9.20%	6.13%
Return on assets	11.14%	7.25%
Return on equity ("ROE")	26.17%	11.13%

Answer 3(b)(i)

Liquidity:

Vertex is in a better liquidity position than Peak to repay its short-term debts as reflected in its higher current ratio and quick ratio.

Answer 3(b)(ii)

Asset management efficiency:

Vertex has higher inventory turnover and total assets turnover ratios and shorter days' receivables than Peak, which imply that Vertex is more efficient in selling its inventories and managing its total assets within a shorter time period to collect the trade receivables from its customers.

Answer 3(b)(iii)

Leverage:

Comparing the two companies, Vertex has more debt financing than Peak, as reflected in its higher debt ratio and equity multiplier. Meanwhile, Vertex has a higher times-interest-earned ratio which implies that Vertex is capable of generating enough profits to fulfil its interest obligations despite its higher financing cost as compared with Peak.

Answer 3(b)(iv)

Profitability:

Vertex performs better than Peak in profitability, as Vertex has significantly higher ratios as measured by net profit margin, return on assets, and return on equity.

Answer 3(c)

DuPont formula

	<u>Vertex</u>	<u>Peak</u>
Net profit margin x	9.20%	6.13%
Total asset turnover x	1.21	1.18
Equity multiplier	2.35	1.53

= Return on Equity

26.17% *11.07%

*Rounding difference of 0.06%

Vertex has a higher ROE.

Vertex has a higher ROE achieved by a higher net profit margin and total asset turnover as a result of better cost control and asset management as compared with Peak.

In addition, Vertex has a higher ROE by leveraging its capital structure as evidenced by a higher equity multiplier.

Vertex employs more debt finance in its capital structure to benefit from the tax shield of the loan interest deductibility; but at the same time, it may also be exposed to more overall financial risks as compared with Peak.

Answer 3(d)

Toppoint should acquire Vertex as it has significantly outstanding performance in liquidity, asset management, leverage, and also profitability.

Answer 4(a)

(i) $R_E \text{ (DGM)} = (\text{HK\$}1.4 \times 1.05) / \text{HK\$}52 + 5\% = 7.83\%$
 $R_E \text{ (CAPM)} = 4\% + 2.3 (15\% - 4\%) = 29.30\%$
 $\Rightarrow R_E = (7.83\% + 29.30\%) / 2 = 18.57\%$

(ii) $R_P = \text{HK\$}100 \times 6\% / \text{HK\$}98 = 6.12\%$

(iii) $R_D = \text{yield to maturity} = 7\%$

Answer 4(b)

$D = 13,000 \times \text{HK\$}1,000 = \text{HK\$}13,000,000$

$P = 25,000 \times \text{HK\$}98 = \text{HK\$}2,450,000$

$E = 120,000 \times \text{HK\$}52 = \text{HK\$}6,240,000$

$V = \text{HK\$}13,000,000 + \text{HK\$}2,450,000 + \text{HK\$}6,240,000 = \text{HK\$}21,690,000$

$W_D = D/V = \text{HK\$}13,000,000 / \text{HK\$}21,690,000 = 59.93\%$

$W_P = P/V = \text{HK\$}2,450,000 / \text{HK\$}21,690,000 = 11.30\%$

$W_E = E/V = \text{HK\$}6,240,000 / \text{HK\$}21,690,000 = 28.77\%$

$WACC = 28.77\% \times 18.57\% + 11.30\% \times 6.12\% + 59.93\% \times 7\% \times (1 - 40\%) = 8.55\%$

Answer 4(c)

$$\text{NPV(A)} = -\text{HK\$}1,680,000 + \text{HK\$}200,000 / 1.0855 + \text{HK\$}350,000 / 1.0855^2 + \text{HK\$}570,000 / 1.0855^3 + \text{HK\$}820,000 / 1.0855^4 = -\text{HK\$}162,474.86$$

$$\text{NPV(B)} = -\text{HK\$}730,000 + \text{HK\$}230,000 / 1.0855 + \text{HK\$}350,000 / 1.0855^2 + \text{HK\$}400,000 / 1.0855^3 + \text{HK\$}510,000 / 1.0855^4 = \text{HK\$}458,975.46$$

The Company should choose Project B.

Answer 4(d)

IRR should be larger than the WACC for Project B (with positive NPV), whereas IRR should be smaller than the WACC for Project A (with negative NPV).

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